

20 a) $(1, 2, 0, 1) \rightarrow P(x) = 1 + 2x + 0x^2 + 1x^3 = 1 + 2x + x^3$

evaluate at $1, 2, \dots, 6 \quad \hat{=} \quad a_i$

$$P(1) = 1 + 2 \cdot 1 + 1^3 = 4 \quad (\text{mod } 11)$$

$$P(2) = 1 + 2 \cdot 2 + 2^3 = 13 = 2$$

$$P(3) = 1$$

$$P(4) = 7$$

$$P(5) = 4$$

$$P(6) = 9$$

$$\text{Encoded message} = (4, 2, 1, 7, 4, 9)$$

b) received $\hat{=}$ $P(u_i)$: $(4, 2, 1) \in (4, 9)$ use these for decoding
set up $g_i(x)$: $g_i(x) = (x-1)(x-2)(x-3)(x-5)$
1 factor will always be discarded

$$g_1(x) = (x-2)(x-3)(x-5)$$

$$g_2(x) = (x-1)(x-3)(x-5)$$

$$g_3(x) = (x-1)(x-2)(x-5)$$

$$g_5(x) = (x-1)(x-2)(x-3)$$

plug-in the used evaluation positions : 1, 2, 3, 5

$$g_1(1) = (1-2)(1-3)(1-5) = 3$$

$$g_2(2) = 3$$

$$g_3(3) = 7$$

$$g_5(5) = 2$$

mod 11!

$$g_1^{-1}(1) = 4$$

$$g_2^{-1}(2) = 4$$

$$g_3^{-1}(3) = 8$$

$$g_5(5) = 6$$

$$\frac{p(u_i)}{g_i(u_i)} = p(u_i) \cdot g_i^{-1}(u_i) \rightarrow$$

$$\begin{array}{rcl} 4 & \cdot & 4 = 5 \\ 2 & \cdot & 4 = 8 \\ 1 & \cdot & 8 = 8 \\ 4 & \cdot & 6 = 2 \end{array}$$

mod 11

$$\begin{aligned} \rightarrow p(x) &= 5g_1(x) + 8g_2(x) + 8g_3(x) + 2g_5(x) \\ &= x^3 + 2x + 1 \end{aligned}$$

$\rightarrow$ original message : $(1, 2, 0, 1)$

$$c) \left\lceil \frac{n-m}{2} \right\rceil = \left\lceil \frac{6-4}{2} \right\rceil = 1$$

$$1+4-1=4$$

$$f(x) = f_0 + f_1 x$$

$$g(x) = g_0 + g_1 x + g_2 x^2 + g_3 x^3 + g_4 x^4$$

$$p(x, y) = y \cdot f(x) + g(x)$$

→ see separate document: $f(x) = 10x + 6$ / $g(x) = x^4 + 5x^3 + 2x^2 + 5$

$$(x^4 + 5x^3 + 2x^2 + 5) : (10x + 6) = 10x^3 + 9x + 10$$

$$-(x^4 + 60x^3)$$

$$\underline{2x^2 + 5}$$

$$-(2x^2 + 54x)$$

$$\underline{x + 5}$$

$$-(x + 60) = 5$$

$$-(10x^3 + 9x + 10) = x^3 + 2x + 1$$

→ original message: (1, 2, 0, 1)